Overview

The announcement asks you to practice the **Goldilocks principle** in mathematical writing: give *just the right amount* of explanation — not too little and not too much. These notes explain what that means, give short course-related examples, a checklist you can use in class or on exams, and practical tips for practicing the skill.

1 What the announcement means (short)

- **Too little:** You omit necessary steps or justification; the reader cannot follow or verify your reasoning.
- **Too much:** You repeat or restate obvious facts, or include irrelevant digressions that bury the main point.
- **Just right:** Provide the definitions you need, the logical steps that connect them, and a clear conclusion — compact, precise, and sufficient for the intended audience to verify the result.

2 Why it matters beyond exams

- *Homework & assignments:* graders need enough to see you understood the idea — but not long-winded prose.
- *Research / papers:* reviewers want clear, rigorous proofs and concise motivation.
- *Coding / code review:* succinct comments and correct edge-case handling beat verbose, noisy comments.
- *Presentations / teaching:* clarity and economy keep the audience engaged.
- *Professional writing:* concise but sufficient detail (design docs, emails, bug reports) leads to faster, less-error-prone collaboration.

3 Concrete course examples

Below are short "too little / too much / just right" style model answers for typical course problems.

3.1 Equivalence relation

Example 3.1 (Relation defined by divisibility)**.** Let R be a relation on $\mathbb{Z}$ defined by

$$a\,R\,b \iff 3 \mid (a-b).$$

Show that R is an equivalence relation.

Just-right solution:

Proof. **Reflexive.** For any $a \in \mathbb{Z}$, $a - a = 0$ and $3 \mid 0$; thus aRa.

Symmetric. If $3 \mid (a-b)$ then $3 \mid (b-a)$, so $aRb \Rightarrow bRa$.

Transitive. If $3 \mid (a-b)$ and $3 \mid (b-c)$, then $3 \mid (a-c)$; hence aRb and $bRc \Rightarrow aRc$.

Therefore R is reflexive, symmetric, and transitive, i.e., an equivalence relation. $\square$

3.2 Function: bijectivity

Example 3.2. Let $f : \mathbb{Z} \to \mathbb{Z}$ be defined by $f(n) = n + 1$. Prove f is bijective.

Just-right solution:

Proof. **Injective.** If $f(a) = f(b)$ then $a + 1 = b + 1$, so $a = b$.

Surjective. For arbitrary $y \in \mathbb{Z}$ choose $x = y - 1$. Then $f(x) = x + 1 = y$.

Thus f is both injective and surjective, hence bijective. □

3.3 Partial order: subset relation

Example 3.3. Show $(\mathcal{P}(X), \subseteq)$ is a partially ordered set.

Just-right solution:

Proof. **Reflexive.** For any $A \subseteq X$, $A \subseteq A$.

Antisymmetric. If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

Transitive. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Hence $\subseteq$ satisfies reflexivity, antisymmetry, and transitivity, so it is a partial order. □

4 A short checklist for "Goldilocks" answers

Before submitting or handing in a solution, run this mental checklist:

1. **State what you are proving** in one crisp sentence.
2. **Give necessary definitions** (only those used in the argument).
3. **Outline the plan** if the proof is longer than a couple of lines.
4. **Do the steps:** each step should follow logically and be minimally justified.
5. **Address edge cases** when relevant (e.g., empty set, domain restrictions).
6. **Conclude clearly:** "Hence ... is ..." or similar.
7. **Edit for redundancy:** remove repeated or obvious statements.

5 Practical tips to practice this skill

- After writing a solution, *try removing one sentence.* If the argument still works, that sentence was probably redundant; if it breaks, put it back.
- Conversely, *remove a justification line* — if the reader cannot fill the gap, restore the line.
- When reviewing classmates' solutions, mark places that are unclear (too little) or repetitive (too much).
- Time yourself: write a focused 5–7 line solution to a typical problem; this trains concise clarity.

One-line summary

Think like a careful editor: include everything the reader needs to verify your claim, and nothing they don't.

If you'd like, I can:

- Convert this to a handout-ready two-column format,
- Generate a short practice worksheet with three problems and model "too little / too much / just right" solutions, or
- Rewrite one of your past solutions into the three-style format so you can compare.

Propositions on Lattices and Chains

The truth values for the following propositions are determined.

1. Every finite lattice has a least element.
   True. By repeatedly taking the greatest lower bound (GLB) of pairs of elements in a finite lattice, one can find a single GLB for the entire set, which is the least element.
2. The poset $(\{1, 2, 4, 8, 16\}, |)$ is a lattice.
   True. This poset is a chain (a totally ordered set), and every chain is a lattice. For any two elements a, b where $a \mid b$, their LUB is b and their GLB is a.
3. If A is a set, the poset $(\mathcal{P}(A), \subseteq)$ is a lattice.
   True. For any two subsets $X, Y \subseteq A$, their least upper bound (LUB) is their union $(X \cup Y)$ and their greatest lower bound (GLB) is their intersection $(X \cap Y)$.
4. There exist an infinite lattice which has a least but not a greatest element.
   True. The set of positive integers with the divisibility relation, $(\mathbb{Z}^+, |)$, is an infinite lattice. It has a least element (1) but no greatest element.
5. If a poset is a lattice, then it is a chain.
   False. A lattice does not require every pair of elements to be comparable. For example, in the lattice $(\mathcal{P}(\{a, b\}), \subseteq)$, the elements $\{a\}$ and $\{b\}$ are incomparable.
6. There exist an infinite lattice which has a greatest but not a least element.
   True. The set of negative integers with the divisibility relation, $(\mathbb{Z}^-, |)$, is a lattice with a greatest element (-1) but no least element.
7. Every finite lattice has a greatest element.
   True. This is the dual to proposition 1. By repeatedly taking the least upper bound (LUB) of elements, one can find a greatest element for the entire finite set.
8.
9. If a poset is a chain, then it is a lattice.
   True. In a chain, any two elements are comparable. If $a \preceq b$, their LUB is b and their GLB is a. Since a LUB and GLB exist for every pair, every chain is a lattice.
10.
11. If a poset is a lattice, then its dual poset is also a lattice.
    True. The LUB in a poset becomes the GLB in its dual, and vice-versa. If LUBs and GLBs exist for all pairs in a poset, they are guaranteed to exist in its dual.
12. Every nonempty finite subset of a lattice has a greatest lower bound.
    True. This property can be proven by induction on the size of the subset, repeatedly applying the pairwise GLB operation which is guaranteed to exist in a lattice.
13. There exist an infinite lattice which has neither a least nor a greatest element.
    True. The set of integers with the standard order, $(\mathbb{Z}, \leq)$, is a chain (and therefore a lattice) with no least and no greatest element.
14.
15. Every nonempty finite subset of a lattice has a least upper bound.
    True. This is the dual to proposition 12 and follows from the same inductive argument using the LUB operation.
16. The poset $(\{1, 2, 3, 4, 5\}, |)$ is a lattice.
    False. A lattice requires a LUB for every pair to exist *within the set*. The pair $\{2, 3\}$ has a least common multiple of 6, which is not in the set.

17. The poset $(\mathbb{Z}^+, |)$ is a lattice.
True. For any two positive integers a, b, their GLB is their greatest common divisor ($\gcd(a, b)$) and their LUB is their least common multiple ($\text{lcm}(a, b)$), both of which always exist.

18. There exist an infinite lattice which has both a least and a greatest element.
True. The power set of the natural numbers, $(\mathcal{P}(\mathbb{N}), \subseteq)$, is an infinite lattice with a least element ($\emptyset$) and a greatest element ($\mathbb{N}$). Another example is the set of real numbers in $[0, 1]$ under $\leq$.

A **lattice** requires that every pair of elements has both:

1. A **greatest lower bound** (meet, $\wedge$)
2. A **least upper bound** (join, $\vee$)

Consider the poset with elements a, b, c, d, e, f as shown in the diagram:

- a is the least element (bottom)
- f is the greatest element (top)
- The middle layer consists of $\{b, c, d, e\}$

Key Issue: Take the pair b and c:

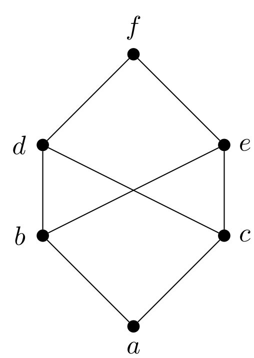
- Their common upper bounds are $\{d, e, f\}$
- The least upper bound should be unique
- However, both d and e are minimal upper bounds of $\{b, c\}$, and neither is below the other:

$$d \not\leq e \quad \text{and} \quad e \not\leq d$$

Thus, the pair $\{b, c\}$ has two minimal upper bounds instead of one. This violates the lattice property, which requires a unique least upper bound for every pair.

Conclusion: The given poset is **not a lattice**.

Consider the poset with elements a, b, c, d, e, f:



Key Observation

Take the pair of elements b and c:

- Their upper bounds are d, e, f.
- The **least upper bound** should be unique.
- However, both d and e are minimal upper bounds, and neither is below the other:

$$d \not\leq e \quad \text{and} \quad e \not\leq d.$$

Thus, the join $b \vee c$ is not well-defined.

Conclusion

Since b and c do not have a unique least upper bound, this poset fails to satisfy the lattice property. Hence, the given poset is **not a lattice**.

In the context of partially ordered sets (posets), a **chain** is a "perfectly orderly" poset where every element is comparable to every other, while a **lattice** is a "well-structured" poset where any two elements have a defined 'least common successor' and 'greatest common predecessor'.

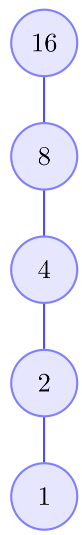
Chains: Total Order

A **chain** is a poset where every pair of elements is comparable. This means for any two elements a and b, either $a \preceq b$ or $b \preceq a$. Because all elements can be placed in a single line of order, this structure is also known as a **totally ordered set**. The Hasse diagram for a chain is always a single vertical path.

Example: Divisors of 16

The poset $(\{1, 2, 4, 8, 16\}, |)$ is a chain because for any two numbers in the set, one divides the other.

Chain: $(\{1, 2, 4, 8, 16\}, |)$



Lattices: Joins and Meets for All

A **lattice** is a poset where every pair of elements has both:

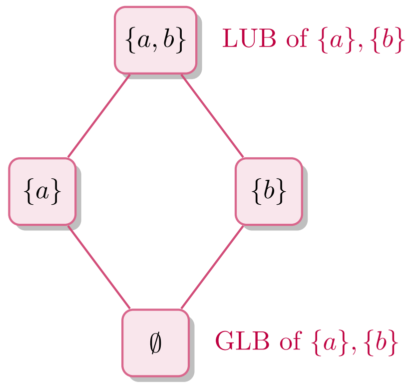
1. A **Least Upper Bound (LUB)**, also called a **join**.
2. A **Greatest Lower Bound (GLB)**, also called a **meet**.

This property guarantees a high degree of structure, ensuring that a unique 'lowest common ancestor' (the LUB) and 'highest common descendant' (the GLB) can be found for any two elements.

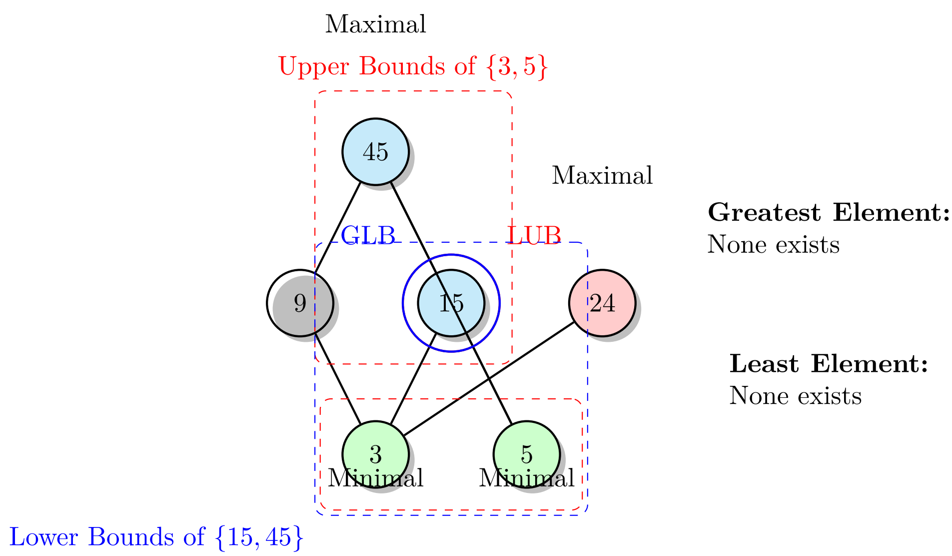
Example: Power Set of $\{a, b\}$

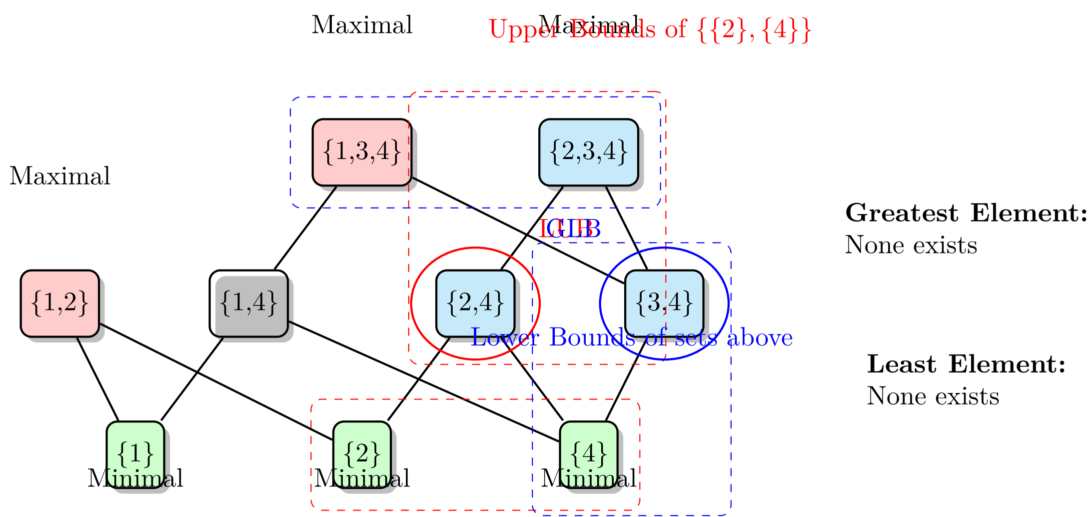
The poset $(\mathcal{P}(\{a, b\}), \subseteq)$ is a classic example of a lattice. For any two subsets in this poset, their **union** is the LUB and their **intersection** is the GLB.

Lattice: $(\mathcal{P}(\{a,b\}), \subseteq)$



Solutions for Poset Questions





Poset $(\{3, 5, 9, 15, 24, 45\}, |)$

For this poset, where the relation is "divides", the following properties are first identified:

- **Minimal Elements** (no other element divides them): $\{3, 5\}$
- **Maximal Elements** (they do not divide any other element in the set): $\{24, 45\}$

1. Select all maximal elements.
An element is maximal if no other element in the set is a multiple of it.
Correct options: **F. 24** and **E. 45**.

2. Select all minimal elements.

An element is minimal if it is not a multiple of any other element in the set.
Correct options: **B. 5** and **C. 3**.

3. Select the greatest element if one exists.
A greatest element must be a unique maximal element. As there are two maximal elements, no greatest element exists.
Correct option: **G. None of the above**.

4. Select the least element if one exists.
A least element must be a unique minimal element. As there are two minimal elements, no least element exists.
Correct option: **G. None of the above**.

5. Select all upper bounds of $\{3, 5\}$.
An upper bound must be a multiple of both 3 and 5. The elements from the set that satisfy this are 15 and 45.
Correct options: **F. 15** and **E. 45**.

6. Select the least upper bound of $\{3, 5\}$ if one exists.
The least upper bound is the minimum of the set of upper bounds $\{15, 45\}$. Since $15 \mid 45$, 15 is the minimum.
Correct option: **C. 15**.

7. Select all lower bounds of $\{15, 45\}$.
A lower bound must divide both 15 and 45. The elements from the set that satisfy this are 3, 5, and 15.
Correct options: **B. 3**, **E. 5**, and **C. 15**.

8. Select the greatest lower bound of $\{15, 45\}$ if one exists.
The greatest lower bound is the maximum of the set of lower bounds $\{3, 5, 15\}$. Since 3 and 5 both divide 15, 15 is the maximum.
Correct option: **A. 15**.

Poset $(\{\{1\}, \{2\}, \{4\}, \{1, 2\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}, \subseteq)$

For this poset, where the relation is "is a subset of", the following properties are identified:

- **Minimal Elements** (no proper subsets in the collection): $\{\{1\}, \{2\}, \{4\}\}$
- **Maximal Elements** (not a proper subset of any other set): $\{\{1, 2\}, \{1, 3, 4\}, \{2, 3, 4\}\}$

1. Select all maximal elements.
Correct options: **H. {1,2}**, **I. {1,3,4}**, and **B. {2,3,4}**.

2. Select all minimal elements.
Correct options: **C. {1}**, **E. {2}**, and **B. {4}**.

3. Select the greatest element if one exists.
There are three maximal elements, not a unique one. Therefore, no greatest element exists.
Correct option: **J. None of the above**.

4. Select the least element if one exists.
There are three minimal elements, not a unique one. Therefore, no least element exists.
Correct option: **J. None of the above**.

5. Select all upper bounds of $\{\{2\}, \{4\}\}$.
An upper bound must contain both 2 and 4. The sets from the collection that satisfy this are $\{2, 4\}$ and $\{2, 3, 4\}$.

Correct options: **B. {2,4}** and **G. {2,3,4}**.

6. Select the least upper bound of $\{\{2\},\{4\}\}$ **if one exists.**
The least upper bound is the minimum of the upper bounds $\{\{2,4\},\{2,3,4\}\}$. Since $\{2,4\} \subseteq \{2,3,4\}$, $\{2,4\}$ is the minimum.
Correct option: **D. {2,4}**.

7. Select all lower bounds of $\{\{1,3,4\},\{2,3,4\}\}$**.**
A lower bound must be a subset of the intersection, which is $\{3,4\}$. The sets from the collection that are subsets of $\{3,4\}$ are $\{4\}$ and $\{3,4\}$.
Correct options: **G. {4}** and **C. {3,4}**.

8. Select the greatest lower bound of $\{\{1,3,4\},\{2,3,4\}\}$ **if one exists.**
The greatest lower bound is the maximum of the lower bounds $\{\{4\},\{3,4\}\}$. Since $\{4\} \subseteq \{3,4\}$, $\{3,4\}$ is the maximum.
Correct option: **D. {3,4}**.

The truth values for the following propositions concerning partially ordered sets are determined. A pair of elements a, b in a poset with relation $\preceq$ is considered **comparable** if either $a \preceq b$ or $b \preceq a$ holds true.

1. If $a, b \in \mathbb{R}$, then a and b are comparable in the poset $(\mathbb{R}, \leq)$.
   True. For any two real numbers a and b, it is always true that either $a \leq b$ or $b \leq a$. This is the definition of a total order.
2. The elements 7 and 12 are comparable in the poset $(\mathbb{Z}^+, |)$.
   False. Neither $7 \mid 12$ nor $12 \mid 7$ is true. Since neither element divides the other, they are incomparable.
3. The elements 10 and 5 are comparable in the poset $(\mathbb{Z}^+, |)$.
   True. While $10 \nmid 5$, it is true that $5 \mid 10$. Since one of the relations holds, they are comparable.
4. The sets $\{1, 2, 3\}$ and $\{2, 3, 4\}$ are comparable in the poset $(\mathcal{P}(\mathbb{Z}), \subseteq)$.
   False. Neither set is a subset of the other, so they are incomparable.
5. The dual poset of $(\mathbb{Z}, |)$ is $(\mathbb{Z}, \nmid)$.
   False. The dual of the relation $a \mid b$ is $b \mid a$. The symbol $\nmid$ denotes the negation of the relation, not its dual.
6. The sets $\{1, 2\}$ and $\{1, 2, 3, 4, 5\}$ are comparable in the poset $(\mathcal{P}(\mathbb{Z}), \subseteq)$.
   True. The set $\{1, 2\}$ is a subset of $\{1, 2, 3, 4, 5\}$, so they are comparable.
7. The dual poset of $(\mathbb{Z}, \leq)$ is $(\mathbb{Z}, >)$.
   False. The dual of the relation $\leq$ is $\geq$. The relation $>$ is not a partial order because it is not reflexive.
8. The sets $\{1, 3, 5\}$ and $\{2, 4\}$ are comparable in the poset $(\mathcal{P}(\mathbb{Z}), \subseteq)$.
   False. The sets are disjoint, so neither is a subset of the other. They are incomparable.
9. The relation $\neq$ is a partial ordering of $\mathbb{Z}$.
   False. A partial order must be reflexive, but $a \neq a$ is always false.
10. The set $\emptyset$ is comparable to every element in the poset $(\mathcal{P}(\mathbb{Z}), \subseteq)$.
    True. The empty set is a subset of every set, so $\emptyset \subseteq A$ is true for any set A, making them comparable.
11. The elements 13 and 39 are comparable in the poset $(\mathbb{Z}^+, |)$.
    True. Since $13 \mid 39$, the elements are comparable.
12. The sets $\{3, 5, 7\}$ and $\{7\}$ are comparable in the poset $(\mathcal{P}(\mathbb{Z}), \subseteq)$.
    True. The set $\{7\}$ is a subset of $\{3, 5, 7\}$, so they are comparable.
13. The element 1 is comparable to all other elements in the poset $(\mathbb{Z}^+, |)$.
    True. The number 1 divides every positive integer ($1 \mid k$ for all $k \in \mathbb{Z}^+$), making it comparable to all other elements.
14. The relation $=$ is both a partial order and an equivalence relation on $\mathbb{R}$.
    True. It is reflexive, antisymmetric, and transitive (a partial order). It is also reflexive, symmetric, and transitive (an equivalence relation).
15. The set $\mathbb{Z}$ is comparable to every element in the poset $(\mathcal{P}(\mathbb{Z}), \subseteq)$.
    True. Every set $A \in \mathcal{P}(\mathbb{Z})$ is by definition a subset of $\mathbb{Z}$, so $A \subseteq \mathbb{Z}$ is always true.
16. The dual poset of $(\mathcal{P}(\{0, 1, 2\}), \subseteq)$ is $(\mathcal{P}(\{0, 1, 2\}), \supseteq)$.
    True. The dual of the relation $A \subseteq B$ is $B \subseteq A$, which is equivalent to $A \supseteq B$.
17. The elements 3 and 3 are comparable in the poset $(\mathbb{Z}^+, |)$.
    True. Because the divisibility relation is reflexive, $3 \mid 3$ is true, which makes them comparable.
18. The elements 6 and 9 are comparable in the poset $(\mathbb{Z}^+, |)$.
    False. Neither $6 \mid 9$ nor $9 \mid 6$ is true. They are incomparable.

First, the six permutations on the set $\{1, 2, 3\}$ are explicitly defined for clarity. The permutation c, which was not explicitly defined in the problem description, is the transposition of 1 and 3.

- $e = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$ (identity)
- $a = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$ (swaps 1 and 2)
- $b = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$ (swaps 2 and 3)
- $c = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$ (swaps 1 and 3)
- $d = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ (cycle $1 \to 2 \to 3 \to 1$)
- $f = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$ (cycle $1 \to 3 \to 2 \to 1$)

Completed Product Table

The completed composition table is presented below. The operation is defined as $(row) \circ (column)$.

$\circ$	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	d	f	b	c
b	b	f	e	d	c	a
c	c	d	f	e	a	b
d	d	c	a	b	f	e
f	f	b	c	a	e	d

Analysis of the Table

What patterns are seen in this table?

- **Identity Element**: The first row and first column are identical to the headers. This shows that e is the identity element, as $e \circ x = x \circ e = x$ for any permutation x.
- **Not Commutative**: The table is not symmetric across its main diagonal. For example, $a \circ b = d$, but $b \circ a = f$. This confirms that the composition of permutations is not a commutative operation.
- **Closure**: Every entry in the table is one of the six original permutations. This means the set is closed under the operation of composition.

How many times does a permutation appear in a given row or column?

Each permutation appears **exactly once** in each row and each column. This is known as the **Latin Square Property** and is a fundamental characteristic of a group's Cayley table.

Does every permutation have an inverse?

Yes. For every permutation x, there exists a unique permutation y (the inverse, x^{-1}) such that $x \circ y = e$ and $y \circ x = e$. These can be found by locating the identity element e in each row and column.

- The permutations a, b, and c are their own inverses: $a \circ a = e$, $b \circ b = e$, $c \circ c = e$.
- The permutations d and f are inverses of each other: $d \circ f = e$ and $f \circ d = e$.

- The identity e is its own inverse: $e \circ e = e$.

Cayley Tables and Properties for $\mathbb{Z}_n$

The completed Cayley tables for addition and multiplication in $\mathbb{Z}_4$ and $\mathbb{Z}_5$ are provided, followed by an analysis of their properties.

Cayley Tables for $\mathbb{Z}_4$

The set of congruence classes is $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$.

Addition Modulo 4

+	[0]	[1]	[2]	[3]
[0]	[0]	[1]	[2]	[3]
[1]	[1]	[2]	[3]	[0]
[2]	[2]	[3]	[0]	[1]
[3]	[3]	[0]	[1]	[2]

Multiplication Modulo 4

·	[0]	[1]	[2]	[3]
[0]	[0]	[0]	[0]	[0]
[1]	[0]	[1]	[2]	[3]
[2]	[0]	[2]	[0]	[2]
[3]	[0]	[3]	[2]	[1]

Cayley Tables for $\mathbb{Z}_5$

The set of congruence classes is $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$.

Addition Modulo 5

+	[0]	[1]	[2]	[3]	[4]
[0]	[0]	[1]	[2]	[3]	[4]
[1]	[1]	[2]	[3]	[4]	[0]
[2]	[2]	[3]	[4]	[0]	[1]
[3]	[3]	[4]	[0]	[1]	[2]
[4]	[4]	[0]	[1]	[2]	[3]

Multiplication Modulo 5

·	[0]	[1]	[2]	[3]	[4]
[0]	[0]	[0]	[0]	[0]	[0]
[1]	[0]	[1]	[2]	[3]	[4]
[2]	[0]	[2]	[4]	[1]	[3]
[3]	[0]	[3]	[1]	[4]	[2]
[4]	[0]	[4]	[3]	[2]	[1]

Patterns and Properties

The following observations are made from the tables.

General Patterns

- **Identity Elements**: For addition, [0] is the identity element. For multiplication, [1] is the identity element.
- **Symmetry**: All four tables are symmetric across the main diagonal.

Frequency of Classes in Rows/Columns

- **Addition**: For both $\mathbb{Z}_4$ and $\mathbb{Z}_5$, every congruence class appears **exactly once** in each row and each column. This property means the tables are **Latin squares**.
- **Multiplication**: The pattern is more complex.
  - In $\mathbb{Z}_4$, the rows/columns for [1] and [3] (classes corresponding to integers relatively prime to 4) contain each class exactly once. However, the row/column for [2] contains only [0] and [2], each appearing twice. This is because 2 is a **zero divisor** ($[2] \cdot [2] = [0]$).
  - In $\mathbb{Z}_5$, since 5 is a **prime number**, all non-zero elements are relatively prime to 5. Consequently, in every row and column corresponding to a non-zero element, each congruence class appears **exactly once**.

Commutativity and Associativity

- **Commutativity**: **Yes**, both operations are commutative. This is visually confirmed by the symmetry of the tables and holds because addition and multiplication of integers are commutative:

$$[a] + [b] = [a + b] = [b + a] = [b] + [a]$$

$$[a] \cdot [b] = [ab] = [ba] = [b] \cdot [a]$$

- **Associativity**: **Yes**, both operations are associative. This property is inherited from the associativity of operations in the integers ($\mathbb{Z}$):

$$([a] + [b]) + [c] = [(a + b) + c] = [a + (b + c)] = [a] + ([b] + [c])$$

$$([a] \cdot [b]) \cdot [c] = [(ab)c] = [a(bc)] = [a] \cdot ([b] \cdot [c])$$

The truth values for the following statements are determined. A statement is considered **true** if it holds for all functions and all subsets, while it is **false** if a single counterexample can be found. For the counterexamples provided, the function $f : \{1, 2, 3\} \to \{a, b, c\}$ defined by $f(1) = a$, $f(2) = b$, and $f(3) = a$ is used. This function is neither one-to-one nor onto.

1. $f(A \cap B) \subseteq f(A) \cap f(B)$
   True. If an element $y \in f(A \cap B)$, then $y = f(x)$ for some $x \in A \cap B$. Since $x \in A$ and $x \in B$, it follows that $f(x) \in f(A)$ and $f(x) \in f(B)$. Thus, $y \in f(A) \cap f(B)$.
2. $f^{-1}(C \cup D) \subseteq f^{-1}(C) \cup f^{-1}(D)$
   True. If $x \in f^{-1}(C \cup D)$, its image $f(x)$ is in $C \cup D$. This means $f(x) \in C$ or $f(x) \in D$, which implies $x \in f^{-1}(C)$ or $x \in f^{-1}(D)$.
3. $f^{-1}(C) \cup f^{-1}(D) \subseteq f^{-1}(C \cup D)$
   True. If $x \in f^{-1}(C) \cup f^{-1}(D)$, then $x \in f^{-1}(C)$ or $x \in f^{-1}(D)$. This means $f(x) \in C$ or $f(x) \in D$. Therefore, $f(x) \in C \cup D$, which means $x \in f^{-1}(C \cup D)$.
4. $f(A) \cap f(B) \subseteq f(A \cap B)$
   False. This can fail if the function is not one-to-one. Let $A = \{1\}$ and $B = \{3\}$. Then $f(A) = \{a\}$ and $f(B) = \{a\}$, so $f(A) \cap f(B) = \{a\}$. However, $A \cap B = \emptyset$, so $f(A \cap B) = \emptyset$. The statement $\{a\} \subseteq \emptyset$ is false.
5. If $C \subseteq D$, then $f^{-1}(C) \subseteq f^{-1}(D)$.
   True. If $C \subseteq D$, then any element x whose image $f(x) \in C$ also has its image in D. By definition, any $x \in f^{-1}(C)$ must also be in $f^{-1}(D)$.
6. If $f^{-1}(C) \subseteq f^{-1}(D)$, then $C \subseteq D$.
   False. This can fail if the function is not onto. Let $C = \{c\}$ and $D = \{a\}$. The premise $f^{-1}(C) \subseteq f^{-1}(D)$ is true because $f^{-1}(C) = \emptyset$ and $f^{-1}(D) = \{1, 3\}$. However, the conclusion $C \subseteq D$ is false.
7. $f^{-1}(C) \cap f^{-1}(D) \subseteq f^{-1}(C \cap D)$
   True. If $x \in f^{-1}(C) \cap f^{-1}(D)$, its image $f(x)$ is in both C and D. Therefore, $f(x) \in C \cap D$, which means $x \in f^{-1}(C \cap D)$.
8. $f\left(f^{-1}(C)\right) \subseteq C$
   True. The set $f^{-1}(C)$ contains only elements that map into C. Therefore, the image of this set must be a subset of C.
9. $f(A) \cup f(B) \subseteq f(A \cup B)$
   True. If $y \in f(A) \cup f(B)$, then y is the image of an element in A or in B. In either case, it is the image of an element from $A \cup B$.
10. $A \subseteq f^{-1}(f(A))$
    True. Every $x \in A$ maps to an element $f(x) \in f(A)$. By the definition of inverse image, x must therefore belong to the set of all elements that map into $f(A)$.
11. $f(A \cup B) \subseteq f(A) \cup f(B)$
    True. If $y \in f(A \cup B)$, it is the image of some $x \in A \cup B$. If $x \in A$, then $y \in f(A)$. If $x \in B$, then $y \in f(B)$. So, y must be in $f(A) \cup f(B)$.
12. $f^{-1}(C \cap D) \subseteq f^{-1}(C) \cap f^{-1}(D)$
    True. If $x \in f^{-1}(C \cap D)$, its image $f(x) \in C \cap D$. This means $f(x) \in C$ and $f(x) \in D$. Therefore, x is in both $f^{-1}(C)$ and $f^{-1}(D)$.
13. $f^{-1}(f(A)) \subseteq A$
    False. This can fail if the function is not one-to-one. Let $A = \{1\}$. Then $f(A) = \{a\}$. The inverse image $f^{-1}(\{a\})$ is $\{1, 3\}$. The statement $\{1, 3\} \subseteq \{1\}$ is false.
14. If $f(A) \subseteq f(B)$, then $A \subseteq B$.
    False. This can fail if the function is not one-to-one. Let $A = \{1\}$ and $B = \{3\}$. Then $f(A) = \{a\}$ and $f(B) = \{a\}$. The premise $f(A) \subseteq f(B)$ is true, but the conclusion $A \subseteq B$ is false.
15. $C \subseteq f\left(f^{-1}(C)\right)$
    False. This can fail if the function is not onto. Let $C = \{c\}$. The inverse image $f^{-1}(\{c\})$ is $\emptyset$. The image $f(\emptyset)$ is also $\emptyset$. The statement $\{c\} \subseteq \emptyset$ is false.

16. If $A \subseteq B$, then $f(A) \subseteq f(B)$.
True. If A is a subset of B, any element in A is also in B. Therefore, the image of any element from A must also be an image of an element from B.

§ Principle of Inclusion-Exclusion (PIE) Let $\mathcal{U}$ be a universal set for a counting problem. Let $C_1, C_2, \ldots, C_n$ be a finite collection of finite sets where each $C_i \subseteq \mathcal{U}$. The intersection of all C_i defines a condition set for the cardinality, C, as follows.

$$C = \bigcap_{i \in [n]} C_i = C_1 \cap C_2 \cap \ldots \cap C_n$$

and the complementary sets $\overline{C_i} = \mathcal{U} \setminus C_i$ for $i \in [n] = \{1, 2, \ldots, n\}$ define the sets of elements that do not satisfy the collection of conditions, C. Then, the number of elements in $\mathcal{U}$ that satisfy all the conditions in C is given as follows.

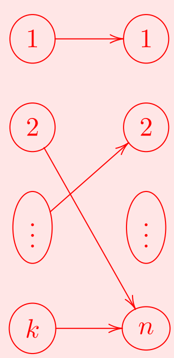
$$\begin{aligned}|C| = \left|\bigcap_{i=1}^{n} C_i\right| &= |U| - \left|\bigcup_{i=1}^{n} \overline{C_i}\right| = \sum_{I \subseteq [n]} (-1)^{|I|} \left|\bigcap_{i \in I} \overline{C_i}\right| \\ &= |U| - \sum_{i=1}^{n} |\overline{C_i}| + \sum_{1 \le i < j \le n} |\overline{C_i} \cap \overline{C_j}| - \ldots + (-1)^n |\overline{C_1} \cap \overline{C_2} \cap \ldots \cap \overline{C_n}|\end{aligned}$$

where the intersection over an empty index set, $\bigcap_{i \in \emptyset} \overline{C_i}$, is defined as the universal set U.

§ Cardinality of Surjective Functions The number of surjective functions from a set $[k] = \{1, 2, \ldots, k\}$ to a set $[n] = \{1, 2, \ldots, n\}$ is given by the formula:

$$\mathcal{S}(k, n) = \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n - i)^k$$

Proof: By the definition of surjective functions, a function $f : [k] \to [n]$ is surjective if and only if for every $i \in [n]$, there exists at least one $j \in [k]$ such that $f(j) = i$. Let C_i be defined as follows.



C_1: there exists a value in $[k]$ that maps to 1 in $[x]$

C_2: there exists a value in $[k]$ that maps to 2 in $[x]$

C_i: there exists a value in $[k]$ that maps to i in $[x]$

Then, $|C|$ is the number of surjective functions by definition. Then, by Principle of Inclusion-Exclusion (PIE), the number of surjective functions is given as follows.

$$\begin{aligned}|C| &= n^k - \binom{n}{1}(n-1)^k + \binom{n}{2}(n-2)^k - \ldots + (-1)^n \binom{n}{n}(n-n)^k \\ &= \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)^k\end{aligned}$$

since $\sum_{i=1}^{n} |\overline{C_i}|$ is the number of functions from $[k]$ to $[n]$ such that a single value, i, is not in the range for each $i \in [n]$, which is given by $\binom{n}{1}(n-1)^k$. With similar reasoning, $\sum_{1 \le i < j \le n} |\overline{C_i} \cap \overline{C_j}| = \binom{n}{2}(n-2)^k$ and so on.

■

Remark: The following definitions (or results) are already proved in class or on previous ICA, Wbwk, Hmwk assignments, and in Sections 4.4, 4.5, and 4.6 of the textbook.

§ Set Equality Let A and B be sets. The sets A and B are equal ($A = B$) iff $A \subseteq B$ and $B \subseteq A$.

§ Subset Relation Let A and B be sets. A is a subset of B ($A \subseteq B$) iff for all $x \in A$ only if $x \in B$.

§ Union of Sets Let A and B be sets. The union of A and B, denoted $A \cup B$, is the set of all elements that are in A, or in B, or in both. That is, $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

§ Intersection of Sets Let A and B be sets. The intersection of A and B, denoted $A \cap B$, is the set of all elements that are in both A and B. That is, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

§ Set Difference Let A, B be subsets of a universal set $\mathcal{U}$. The set difference $A \setminus B$ is defined to be the set $\{x \in \mathcal{U} \mid x \in A \text{ and } x \notin B\}$.

It is sometimes called the relative complement of B in A, and denoted in the textbook as $A - B$.

§ Power Set Let A be a set. The power set of A, denoted $\mathcal{P}(A)$ or $2^{|A|}$, is the set of all subsets of A. That is, $\mathcal{P}(A) = \{S \in \mathcal{U} : S \subseteq A\}$.

§ Complementary Set Let A be a subset of the universal set $\mathcal{U}$. The complement of A, denoted A^c or $\overline{A}$, is the set of all elements in $\mathcal{U}$ that are not in A. That is, $\overline{A} = \{x \in \mathcal{U} : x \notin A\}$.

§ Cartesian Product Let A and B be sets. The Cartesian product of A and B, denoted $A \times B$, is the set consisting of all ordered pairs:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

§ Pairwise Disjoint Sets Let $A_1, A_2, \ldots, A_n$ be sets. These sets are **pairwise disjoint** if for all $i \neq j$, we have $A_i \cap A_j = \emptyset$.

Remark: The following definitions (or results) are already proved in class or on previous ICA, Wbwk, Hmwk assignments, and contents of the textbook.

§ Binary Relation Let A and B be sets. A binary relation R from A to B is a subset of $A \times B$. If (and only if)

$$(a, b) \in R \subseteq A \times B$$

, then we say that a is related to b and write $a \, R \, b$. Otherwise, $a \not R \, b$.

§ Digraph A digraph (or directed graph) is a set of vertices V together with a set of edges $E \subseteq V \times V$.

§ Bipartite Digraph A digraph is bipartite if (and only if) there exist subsets $A, B \subseteq V$ such that

1. $V = A \cup B$ where $A \cap B = \emptyset$ and $A, B \neq \emptyset$.
2. for every $e \in E, e = (a, b)$ for some $a \in A, b \in B$.

Example: Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Then $R = \{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B. Binary relations on finite sets like this can be visualized as a bipartite digraph:

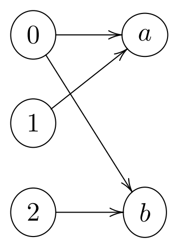


Figure 1: A bipartite graph example

§ Inverse Relation Let A and B be sets. Let R be a relation from A to B. The inverse relation R^{-1} from B to A is the set of ordered pairs $\{(b, a) \mid (a, b) \in R\}$.

§ Relation on a Set A relation on a set $\boldsymbol{A}$ is a relation from A to A.

§ Relation Properties Let A be a set and R a relation on A. Relations on a set are classified by their properties. The most important properties are:

1. **Reflexive**: The relation R is reflexive if (and only if) for all $x \in A$, $(x, x) \in R$.
2. **Symmetric**: The relation R is symmetric if (and only if) for all $x, y \in A$, $(x, y) \in R$ implies $(y, x) \in R$.
3. **Antisymmetric**: The relation R is antisymmetric if (and only if) for all $x, y \in A$, $(x, y) \in R$ and $(y, x) \in R$ only if $x = y$.
4. **Asymmetric**: The relation R is asymmetric if (and only if) for all $x, y \in A$, if $(x, y) \in R$, then $(y, x) \notin R$.
5. **Transitive**: The relation R is transitive if (and only if) for all $x, y, z \in A$, if $(x, y) \in R$ and $(y, z) \in R$, then $(x, z) \in R$.

§ Equivalence Classes Form Partition Let R be an equivalence relation on a nonempty set A. Then $P = \{[a]_R \mid a \in A\}$, the set of equivalence classes resulting from R, is a partition of A.

§ Partition Induces Equivalence Relation Let A be a nonempty set and $P = \{S_i \mid i \in I\}$ a partition of A. Then there exists an equivalence relation R which has the sets $S_i, i \in I$, as its equivalence classes.

Remark: Note that the set I above is an arbitrary index set. It is not necessarily finite, countable, etc!

§ Modulo Operation Let $a, m \in \mathbb{Z}$ with m positive. The remainder when a is divided by m is denoted $\boldsymbol{a}$ mod $\boldsymbol{m}$.

§ Congruence Modulo Let $a, b, m \in \mathbb{Z}$ with m positive. If $m \mid (a-b)$ then we say that $\boldsymbol{a}$ is congruent to $\boldsymbol{b}$ modulo $\boldsymbol{m}$ and write $\boldsymbol{a} \equiv \boldsymbol{b} (\bmod \boldsymbol{m})$.

Example: Let R be an equivalence relation on the set $\{1,2,3,4,5,6,7\}$ such that $1R3, 3R4, 4R7$, and $2R6$. Suppose that R has three equivalence classes.

Example: Let S be another equivalence relation on the set $\{1,2,3,4,5,6\}$. Suppose the equivalence classes of S are $\{1,4,5\}, \{2,6\}$, and $\{3\}$.

Example: Let $[a]_n$ denote the congruence class of $a \in \mathbb{Z}$ modulo n (that is, the equivalence class of a under the relation congruence modulo n) for integers $n \geq 2$. Are the following propositions true or false?

§ Equivalence Relation Let A be a nonempty set. A relation R on A is an equivalence relation if and only if R is reflexive, symmetric, and transitive.

§ Partition Let A be a nonempty set. A partition of A is a collection of nonempty, pairwise disjoint subsets of A whose union is A.

§ Partition Refinement Let A be a nonempty set. A partition P_1 of A is a refinement of a partition P_2 of A if and only if every set in P_1 is a subset of some set in P_2.

§ Equivalence Class Let A be a nonempty set and R an equivalence relation on A. For each $a \in A$, the equivalence class of a with respect to R is the set $[a]_R = \{b \in A \mid aRb\}$.

§ Irreflexive Relation Let A be a nonempty set. A relation R on A is irreflexive if and only if for every $a \in A$, $(a, a) \notin R$.

§ Diagonal Relation Let A be a nonempty set. The diagonal relation on A is the set $\Delta = \{(a, a) \mid a \in A\}$.